

Computer Networks Exam

June 24th, 2025

Closed book questions

1.1 Layering

- A. Draw OSI/TCP/IP and Tanenbaum stacks.
- B. Provide one protocol for every Tanenbaum layer.

1.2 DNS

- A. Provide an annotated diagram showing the operation of a DNS system.
- B. Highlight and name the two major phases.
- C. Why is the combination of the two better than applying either end-to-end?
- D. List the reasons that DNS uses UDP rather than TCP.

2.1 TCP

- A. Explain: how are TCP sequences calculated?
- B. Explain the operation of TCP flow control. Draw a diagram you wish.
- C. Define: “window probe”. What does it do and when is it sent?
- D. Explain how TCP flow control interacts with congestion control in case that the two windows differ.

2.2 Transport Layer Support for *Gnutella 0.4*

- A. Which transport layer protocol does it use: TCP or UDP? Why do you believe this choice was made?
- B. If you answered UDP/TCP for part A, what would be the advantages and disadvantages of moving Gnutella to TCP/UDP?

Open book questions

3.1 LoRa Performance Analysis

This question contained an introductory explanation of LoRa. I don't recall what details were given, but you can probably search this up yourself.

- A. In what cases would the LoRa protocol perform best?
- B. When poorly? Provide a detailed justification.
- C. What level of utilization would you expect the LoRa MAC to achieve? Your answer may be as approximate or specific as you wish.

3.2 LoRa Proposal for Improvement

Propose an improved version of the LoRa MAC protocol using the techniques covered in the data link layer. Show how your design would address the problems that you highlighted in your answer to 3.1.

IP Addressing

4.1 IPv4

- A. What type is 172.16.0.1?
- B. What is the broadcast address of 172.29.206.0 (255.254.0.0)?
- C. What is the broadcast address of 172.23.193.64 (255.192.0.0)?
- D. What is the first valid host of subnet 172.22.7.16/28?
- E. What is the last valid host?

4.2 IPv6

- A. Shorten 2041:0000:140f:0000:0000:0000:875b:131b.
- B. What sort of devices should this be assigned to?
- C. Is this likely to be from stateless auto-configuration?
- D. Describe the privacy implications of using stateless auto-configuration.

Routing

Consider a multi-hop mesh network consisting of resource- and energy-constrained devices. The devices communicate using either Zigbee or BLE and are deployed throughout a physical structure. The network has an IPv6 gateway that provides connectivity to external networks.

- A. Which routing protocol from the lectures fits this case best?
- B. Suggest modifications to make it more efficient.